Project MarginSpring: D2 Data Audit Report

Data acquisition and preparation · Wipro utilisation and IT services operating margin · with Appendix A: Datasheet

Item	Detail
Course / milestone	Analytics Capstone Project (ANCP), Term 5, 2026 · D2: Data audit report, datasheet and versioned dataset
Team	Team J · Abhiroop Ghosh, James Mathew, Ashutosh Mishra · Faculty guide: Nikhil D Jonathan
D2 lead	James Mathew
Dataset version	v1.3 · quarterly_panel_v1.3.csv and MarginSpring_D2_Dataset_v1.3.xlsx
Dataset location	Google Drive folder: https://drive.google.com/drive/folders/1jgL-yZ68I4pia01JHxVYuuy4fZpBzGjK?usp=sharing
Audit date	23 October 2026
Coverage	6 Indian IT services firms · 134 firm-quarters · Q1 FY15 (Apr–Jun 2014) to Q2 FY27 (Jul–Sep 2026)

1. Executive summary

Project MarginSpring estimates how many utilisation points Wipro must regain to lift its IT services operating margin from 16.5% back to management's 17.0–17.5% band (D1 charter). It measures what one utilisation point is worth in margin across Indian IT firms, and tests whether TCS's June 2025 bench cap delivered such a gain. D2 builds and audits the data for that work.

The result is a quarterly panel of **134 firm-quarters across six firms** with **718 sourced values**, plus three control series for modelling, 28 management margin-walk statements and a TCS commentary log, all traced to **120 logged sources**. Every value comes from a document the company or a public body published; nothing is imputed, estimated or proxied.

The data is close to complete. **Wipro, the subject of the project, is 100% complete on every field for all 50 quarters.** Across the six firms the panel captures **498 of the 508 core values the firms publish (98%)** for margin, utilisation, headcount and attrition. The one systematic gap is utilisation at TCS and HCLTech, which neither firm discloses.

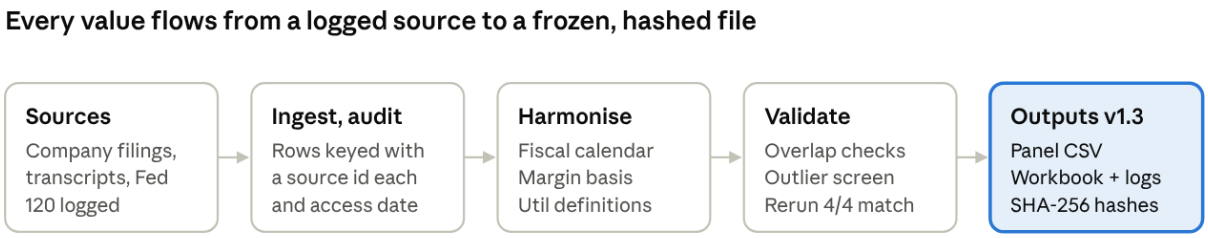


Figure 1. D2 data pipeline: company documents enter with a source id on every row, are harmonised and checked, and leave as hashed files.

Audit item	Result
Firm-quarters in panel	134 (Wipro 50, Infosys 30, TCS 19, LTIMindtree 15, Tech Mahindra 11, HCLTech 9)
Sourced values in panel	718, plus 153 monthly USD/INR rates, 17 salary-hike events, 28 margin walks
Completeness: Wipro / all firms (published core values)	100% / 98% (498 of 508)
Rows traceable to a logged source	134 of 134 (100%)
Duplicate firm-quarter keys	0
Restatements found and logged	2 (Wipro Q4 FY16, Q4 FY18)

Audit item	Result
Outliers flagged / explained	2 / 2 (retained)
Reproducibility rerun	4 of 4 output CSVs byte-identical

1.1 Alignment with the D1 charter

D1 commitment	D2 finding
Six firms, about 40 quarters each	Wipro 50 quarters (Q1 FY15–Q2 FY27), Infosys 30; peers 9–19, limited by their own disclosure history
≥ 15 margin-walk statements across ≥ 4 firms	**Not met.** 6 statements isolate utilisation's effect in bps, across 3 firms; 8 more name it with other drivers. Section 6 asks the guide to make the regression the primary estimate.
≥ 90% of firm-quarters harmonised for utilisation (firms that report it)	**Met:** 98% for Wipro, Infosys and LTIMindtree
100% of inputs traceable to a published source	**Met.** Every row carries a source id listed in the source log.
Each figure cross-checked against two sources	Met for most quarters through overlapping fact sheets and filings; single-source rows flagged by source_type
Controls for wage-hike quarters and one-offs	Salary-hike timing sourced FY21–FY27; one-offs flagged; quarterly USD/INR added
100% public data, no proxies	**Met.**

2. Data audit

2.1 Scope and structure

The unit of analysis is one firm in one Indian fiscal quarter (April–March year; Q2 FY27 = Jul–Sep 2026). Each row of `quarterly_panel_v1.3.csv` holds the firm's reported figures and a pointer to its source. The full field-by-field dictionary is in Appendix A.9.

Field group	Content	Validation
Key: firm, fiscal_quarter, quarter_end	Firm and quarter	No blanks; no duplicate keys
Margin: op_margin_pct, margin_basis	Operating (EBIT) margin and the basis it is reported on	Present in all 134 rows
Capacity: utilisation (2 columns), headcount, attrition	Utilisation excl. and incl. trainees kept separate	Blank only where unpublished
Growth: revenue US$ mn, constant-currency QoQ	Context and controls	Wipro complete; peers partial
Provenance: source_id, source_type, note	Source pointer, document class, caveats	100% of ids in the source log
Controls (workbook): wage_hike_qtr, usd_inr_qavg, one_off_flag	Built by formula from Wage_Hikes, FX_USDINR and notes	660 formulas, 0 errors

2.2 Sources and access

Sources rank in a fixed order of precedence: a company document always overrides a secondary source.

Source family	What it supplied	Licence and terms
Company datasheets, fact sheets, investor releases	Margin, utilisation, headcount, attrition, revenue for all six firms	Public investor disclosure; academic use
Regulatory filings (SEC 6-K, NSE, BSE)	Copies of fact sheets, presentations and transcripts	Public filings
Earnings-call transcripts	Margin walks; TCS management commentary	Quoted sparingly

Source family	What it supplied	Licence and terms
Broker research and news	Gap-filling for LTIMindtree and Tech Mahindra; salary-hike dates	Quote only; marked secondary
Official statistics	Monthly USD/INR (US Federal Reserve G.5 via FRED)	Public domain

All sources were accessed on 23 October 2026, recorded per source in `Source_Log`. Bloomberg terminal exports were also reviewed as an optional cross-check; they were saved without refreshed values and were not needed, since the panel already holds a company source for every value. Their file review is kept in the workbook for reference.

2.3 Completeness by firm and field

Gaps are almost entirely values a firm does not publish, not values that were missed.

Firm	Quarters	Margin	Utilisation (excl. trainees)	Headcount	Attrition
Wipro	50	50	50	50	50
Infosys	30	30	30	29	27 (3 only annualised)
TCS	19	19	not published	19	19
HCLTech	9	9	not published	9	9
LTIMindtree	15	15	13	13	14
Tech Mahindra	11	11	10	11	11
Published values captured		**134 / 134**	**103 / 106**	**131 / 134**	**130 / 134**

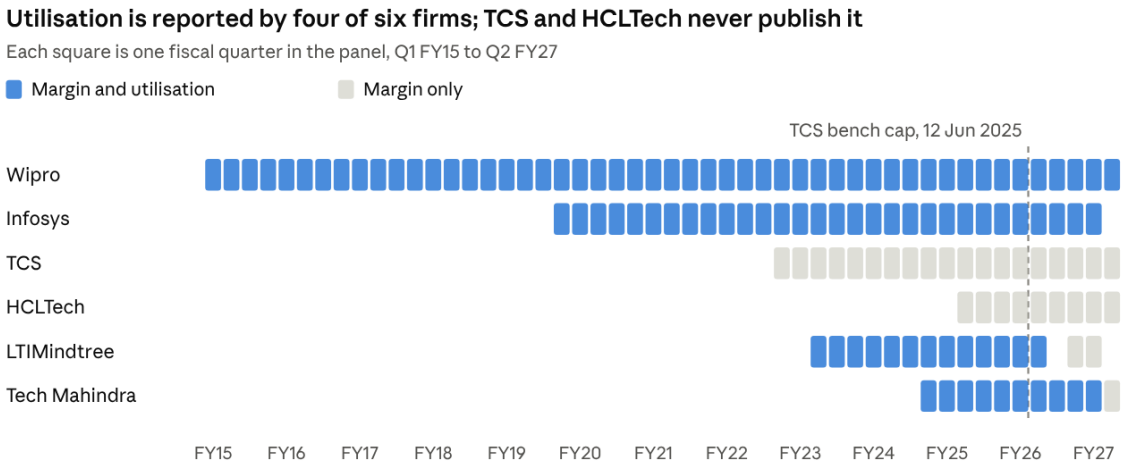


Figure 2. Panel coverage by firm and quarter. Wipro's series runs all 50 quarters; TCS has 13 quarters before its June 2025 bench cap and 5 after.

2.4 Quality issues and treatment

Issue	Evidence	Treatment and rationale
Restatements	Wipro Q4 FY16 margin 20.1% → 19.7% and Q4 FY18 14.4% → 14.6% in the next year's datasheet	Originally reported values kept (D1 rule); restated values in notes
Definition breaks	Wipro headcount basis changes FY16→FY17; TCS reports 'voluntary' LTM attrition from Q2 FY26	No levels compared across breaks; basis noted per row
Conflicting values	LTIMindtree Q4 FY26 15.1% vs 15.5%; HCLTech Mar'25 18.0% vs 17.9%; Infosys Q4 FY26 20.9% vs 21.0%	Value consistent with neighbouring quarters kept; logged
Label conflict	A broker labels LTIMindtree utilisation 'incl. trainees'; values equal the company's 'excl. trainees' figures	Treated as excl. trainees; noted

Issue	Evidence	Treatment and rationale
One-offs	Wipro FY18–FY19 insolvency and impairments, Q2 FY26 provision; HCLTech restructuring 55–122 bps; Labour Codes charges Q3 FY26	Reported value kept; one_off_flag = 1 on 14 rows
Missing values	10 published core values not captured; utilisation unpublished at TCS and HCLTech	Blank, never zero; non-disclosing firms excluded from the utilisation model
Mixed bases	Infosys attrition before FY22 is annualised, not LTM	Excluded from pooled attrition analysis
Outliers	QoQ changes beyond 2.5 s.d. within firm: Wipro Q1 FY19 (+3.2 pts), Q3 FY19 (+4.8 pts)	Rebounds after one-off quarters, confirmed in source; retained

2.5 Harmonisation

- **Margin basis.** Wipro reports an IT Services segment margin; peers report company operating or EBIT margins (HCLTech's includes restructuring). Cross-firm work compares quarter-on-quarter changes in bps, not levels.
- **Utilisation.** Only the excluding-trainees series enters pooled models; other definitions sit in a separate column.
- **Calendar and currency.** All firms use an April–March year. Margins and utilisation are ratios, so no conversion is needed; USD/INR enters only as a control, averaged per fiscal quarter.

2.6 Representativeness and bias

Potential bias	Why it matters	Control
Firm coverage	Six large Indian IT firms, not the whole sector	Results stated for large-cap Indian IT services only
Disclosure	Only four firms publish utilisation	Utilisation model uses those four; TCS and HCLTech enter margin comparisons only
Evidence concentration	Most margin walks come from Infosys	Walks are a cross-check, not the main estimate
Reporting incentive	Firms may cite utilisation more when it helped margin	Positive and negative walks counted separately
Period effects	FY21–FY23 post-COVID demand spike and record attrition	Trend, wage-hike and one-off controls; robustness run excluding FY21–FY23

2.7 Features and leakage control

Planned D4 features: utilisation level and quarterly change, salary-hike quarter flag, one-off flag, quarterly USD/INR, constant-currency growth and a time trend. The target is the reported operating margin; there are no outcome labels. Models are validated on later quarters held out in time order, never on shuffled rows, and every feature uses only the same or earlier quarters. Wipro quarters after its October 2026 bench-policy change (Q3 FY27 onward) are reserved for D6 monitoring and never used for fitting.

2.8 Privacy, consent and responsible use

The data is company-level financial and operating metrics. It holds no employee, client or personal data, so no consent is needed and the DPDP Act, 2023 is not engaged. The only people named are company officers quoted from public calls in their professional roles. Outputs inform a margin-recovery discussion; they are not investment advice and say nothing about any individual employee on the bench.

3. Versioned dataset package

The v1.3 dataset workbook and supporting files are shared in the team Google Drive folder: https://drive.google.com/drive/folders/1jgL-yZ68I4pia01JHxVYuuy4fZpBzGjK?usp=sharing

File	Role
data/processed/quarterly_panel_v1.3.csv	Frozen main table (134 rows, 14 fields)

File	Role
data/processed/ MarginSpring_D2_Dataset_v1.3.xlsx	Working workbook, 15 sheets: panel with live control formulas, source log, harmonisation, quality log, coverage check, wage hikes, USD/INR, TCS commentary
margin_walks_v1.3.csv, source_log_v1.3.csv, policy_events_v1.3.csv	Flat copies of the supporting tables
scripts/panel_data.py, panel_data_v12.py, build_d2_workbook.py	Keyed source data and the build script that recreates every output
README.md	Layout, run order and data rules

Version	Date	Change
v1.3	23 Oct 2026	TCS attrition completed for all 19 quarters
v1.2	23 Oct 2026	Wipro back-filled to FY15; Infosys to FY19; peers extended; salary-hike, USD/INR and TCS commentary tables; 12 more margin walks
v1.1	23 Oct 2026	Peer panel built from company filings
v1.0	23 Oct 2026	Wipro FY20–FY27; first peer rows and margin walks

Rule: any change to content or cleaning logic raises the version; earlier versions are kept.

File	SHA-256
quarterly_panel_v1.3.csv	23d45ebba460c784516737edb435c108c5c376dd260a1f5fbd34a842bb9eacca
margin_walks_v1.3.csv	561fdb0345edf660951f4a1f396b6e70a6e5bda8fd14290093bfde73f573936e
source_log_v1.3.csv	674e89780d008ea1e3e58f6bceb2f721821f457b7354c0873aa1cb13ce9e85cf
policy_events_v1.3.csv	e5262e4c9ec3bd56af95839b982bd583b8106f8a6ba6bbc5ddb2f6d8e1277a0f

Reproducibility. The build was rerun from scratch into an empty folder and all four CSVs matched the frozen copies byte for byte (verify with `sha256sum <file>` or PowerShell `Get-FileHash`). A rerun by a second team member is pending before D3.

4. Validation summary

Check	Result
Rows / firms / duplicate keys	134 / 6 / 0
Wipro quarters, continuous	50 (Q1 FY15–Q2 FY27, no gaps)
Overlapping fact-sheet cross-checks	All matched except 3 logged conflicts of 0.1–0.4 pts
Management statements vs fact sheets	Matched (e.g. Infosys utilisation +0.3, +0.7, +1.9 pts; the D1 ‘+2.4 pts’ walk located in Q2 FY21)
Outlier screen	2 rows flagged; both explained and retained
Workbook formulas	660 formulas, 0 errors

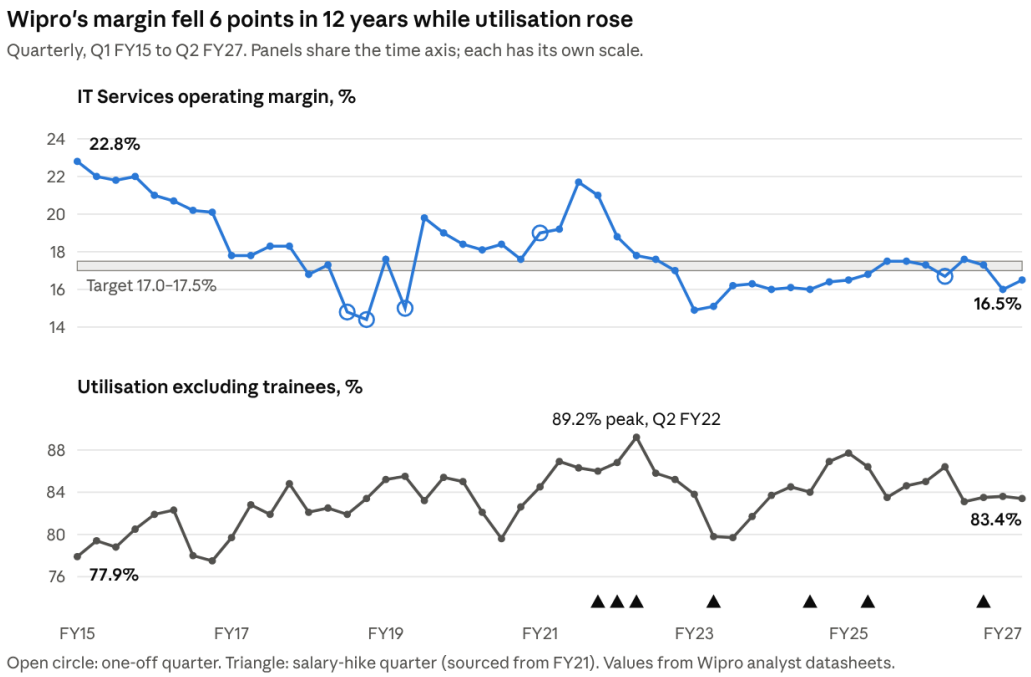


Figure 3. Wipro margin and utilisation, 50 quarters. Raw levels move in opposite directions, which is why D4 needs the trend, salary-hike and one-off controls built in D2.

5. Limitations

- **Undisclosed utilisation.** TCS and HCLTech never publish it, so the TCS bench-cap test compares margin, headcount and attrition; management commentary gives direction, not size.
- **Thin margin-walk evidence.** 6 isolating statements against a target of 15; the computable values (17, 33 and 88 bps per point) are too far apart to set a range alone.
- **Unequal history and controls.** Wipro has 50 quarters, peers 9–30; salary-hike timing is sourced only for FY21–FY27.
- **Definitions and sources.** Margin bases differ by firm; some LTIMindtree and Tech Mahindra rows come from broker tables (matching company documents where they overlap); USD/INR is the Federal Reserve series, not RBI's.
- **Reproduction.** The rerun was scripted on one machine; a second member's rerun is pending.

6. Conclusion and next step

D2 delivers a traceable, versioned and reproducible dataset that covers every later milestone: Wipro complete for 12 years and peers as far as their disclosures allow. A first-pass regression with salary-hike, one-off, USD/INR and trend controls puts one utilisation point at about 15–26 bps of margin for Wipro and about 48 for Infosys; these are early checks, not D4 results. Decisions requested from the guide before D3:

1. Make the regression the primary estimate, with margin walks as the cross-check, because only 6 clean walk statements exist.
2. Run the TCS bench-cap test on margin, headcount and attrition against peers, naming the July 2025 layoffs as a confounder, since TCS does not disclose utilisation.

Next step: D3 exploratory analysis (James Mathew), from `quarterly_panel_v1.3.csv`.

Appendix A. Datasheet: MarginSpring D2 Dataset v1.3

Structured after Gebru et al. (2021), “Datasheets for datasets”, Communications of the ACM 64(12), 86–92. Each answer refers to dataset version v1.3.

A.1 Motivation

Question	Answer
For what purpose was the dataset created?	To estimate the operating-margin value of one utilisation point at Indian IT services firms, derive the utilisation uplift Wipro needs to return to a 17.0–17.5% margin, and test whether TCS’s 2025 bench cap changed margin or headcount.
Who created it, and on whose behalf?	Team J (Abhiroop Ghosh, James Mathew, Ashutosh Mishra), XIME Bangalore, for the ANCP capstone. D2 lead: James Mathew.
Who funded it?	No funding or sponsor. Public sources only.

A.2 Composition

Question	Answer
What do the instances represent?	One firm in one Indian fiscal quarter, with its reported operating metrics.
How many instances are there?	134 firm-quarters: Wipro 50, Infosys 30, TCS 19, LTIMindtree 15, Tech Mahindra 11, HCLTech 9. 718 sourced values in the main table.
Is it a sample or the full population?	A purposive sample: the six largest Indian IT services firms named in D1. Within each firm, every quarter in its range is included.
What data does each instance hold?	14 recorded fields (Appendix A.9) and 5 derived columns in the workbook.
What period does each firm cover?	Wipro Q1 FY15–Q2 FY27; Infosys Q4 FY19–Q1 FY27; TCS Q4 FY22–Q2 FY27; LTIMindtree Q2 FY23–Q1 FY27 (Q3 FY26 missing); Tech Mahindra Q4 FY24–Q2 FY27; HCLTech Q2 FY25–Q2 FY27. Peer start dates follow the earliest quarter for which the firm’s own documents could be retrieved.
Which supporting tables come with it?	Margin_Walks (28 management statements, tiered A: utilisation effect isolated in bps; B: bundled with other drivers; C: named only), Wage_Hikes (17 events), FX_USDINR (153 months), TCS_Commentary (12 statements), Policy_Events (4), Source_Log (120 sources), Data_Quality_Log (19 issues).
Is there a label or target?	No labels. The modelling target is the reported operating margin.
Is any information missing?	Utilisation is not published by TCS or HCLTech. Ten other published core values were not captured (Infosys headcount 1, attrition 3; LTIMindtree utilisation 2, headcount 2, attrition 1; Tech Mahindra utilisation 1). Missing values are blank, never zero.
Are relationships between instances explicit?	Rows link by firm and by quarter; margin walks, policy events, salary hikes and commentary link to panel rows by firm and fiscal quarter.
Are there recommended data splits?	Time-ordered only: fit on earlier quarters, test on the latest 6–8. Wipro Q3 FY27 onward is reserved for D6 monitoring.
Errors, noise or redundancy?	Three small conflicts between sources (0.1–0.4 pts) and two restatements, all logged. Duplicate keys: none.
Is it self-contained?	Yes. Every value is in the files; URLs in the source log allow re-checking against the original documents.
Confidential or sensitive data?	None. No personal data; all values are public company disclosures or public statistics.

A.3 Collection process

Question	Answer
How was the data acquired?	Values were read directly from company PDFs and filings and keyed with a source id per row. Transcripts and broker notes were used for margin walks and gap-filling, and are labelled as such.

Question	Answer
What mechanisms were used?	Company investor-relations sites, SEC EDGAR, NSE and BSE archives, the US Federal Reserve (via FRED), and Python scripts that build and audit the outputs.
Over what timeframe?	Data covers Q1 FY15–Q2 FY27 (Apr 2014–Sep 2026); collected 22–23 October 2026.
Who was involved?	Team J members; no crowdworkers or paid annotators.
Ethical review?	Not required: no human subjects or personal data.

A.4 Preprocessing and cleaning

Question	Answer
What preprocessing was done?	Fiscal labels standardised; originally reported values kept over restatements; estimates excluded; margin basis and utilisation definition recorded per row; quarter-on-quarter changes computed only between consecutive quarters; controls built by formula.
Which rules decided between values?	Precedence: company document > regulatory filing > transcript > broker or news. Original over restated. Actual over estimate. Where two company documents disagree, the value consistent with neighbouring quarters is kept and the other logged.
How were outliers handled?	Quarter-on-quarter changes beyond 2.5 standard deviations within a firm were checked against the source. Both flagged rows (Wipro Q1 FY19, Q3 FY19) were genuine rebounds and kept.
How were missing values handled?	Left blank. No interpolation, carry-forward or imputation, because the D1 charter allows only reported figures.
Was raw data kept?	Yes. Source URLs are logged for every value; Bloomberg reference exports are kept unchanged with SHA-256 hashes.
Is the cleaning code available?	Yes: scripts/build_d2_workbook.py with panel_data.py and panel_data_v12.py in the repository. A clean rerun reproduces the four CSVs byte for byte.

A.5 Uses

Question	Answer
Intended uses	Exploratory analysis of utilisation and margin (D3); margin per utilisation point (D4); Wipro's required uplift (D5); monitoring after Wipro's policy change (D6).
Uses to avoid	Valuing or trading shares; judging individual employees or teams; representing the whole IT sector; comparing margin levels across different margin bases; treating missing utilisation as zero.
Risks to downstream users	Small sample per firm (9–50 quarters) gives wide estimates; report ranges. The margin effect of utilisation can be masked by long-run margin decline unless trend, wage-hike and one-off controls are used.

A.6 Distribution

Question	Answer
How is it distributed?	Shared with the faculty guide through the team Google Drive folder (https://drive.google.com/drive/folders/1jgL-yZ68l4pia01JHxVYuuy4fZpBzGjK?usp=sharing) and kept in a private team repository. Not published openly.
Licence or terms	Company disclosures and filings are public and used for non-commercial academic analysis; transcripts and broker notes are quoted sparingly. Bloomberg reference files stay private under XIME's licence.

A.7 Maintenance

Question	Answer
Who maintains it?	James Mathew (D2 lead).

Question	Answer
Will it be updated?	Yes. v1.4 adds Infosys Q2 FY27 and Wipro Q3 FY27, the first quarter after Wipro's bench-policy change, due about January 2027.
How are versions handled?	Every change raises the version and is logged in the workbook Changelog; earlier versions are never overwritten. Checksums identify each frozen CSV.
How can others contribute or report errors?	Through the team repository; each correction cites a source and is logged in Data_Quality_Log.

A.8 Source register summary

120 sources are logged with publisher, document, URL, periods covered, licence, use and access date. Of the 134 panel rows, 118 (88%) take their values from a company datasheet, fact sheet, release or presentation; 12 from broker tables (LTIMindtree, Tech Mahindra gap-filling); 4 from news or transcripts.

Source type	Sources	Main use
Company disclosure (datasheets, fact sheets, releases, website)	45	Panel values for all six firms
Regulatory filings (SEC 6-K, NSE, BSE)	16	Archived copies of fact sheets and presentations
Earnings-call transcripts and summaries	22	Margin walks, TCS commentary
News, incl. quotes of company officers	26	Salary-hike dates, policy events, gap checks
Broker research	7	Gap-filling for LTIMindtree, Tech Mahindra
Official statistics (Federal Reserve G.5)	1	USD/INR control
Other (teaching case; Bloomberg reference exports)	2	Context; optional cross-check, not used for values
Total	**120**	

A.9 Data dictionary (quarterly_panel_v1.3.csv and workbook columns)

Field	Unit	Definition and notes
firm	text	Wipro, Infosys, TCS, HCLTech, LTIMindtree, Tech Mahindra
fiscal_quarter	text	Indian fiscal quarter, e.g. Q2 FY27 = Jul–Sep 2026
quarter_end	date	Last day of the quarter (YYYY-MM-DD)
op_margin_pct	%	Operating / EBIT margin as reported; see margin_basis
margin_basis	text	Wipro: IT Services segment; Infosys, TCS: company operating margin; HCLTech: company EBIT incl. restructuring; LTIMindtree, Tech Mahindra: company EBIT
util_excl_trainees_pct	%	Utilisation excluding trainees (primary capacity variable)
util_gross_or_incl_trainees_pct	%	Utilisation including trainees, or Wipro gross utilisation; robustness only
headcount	persons	Closing total employees
vol_attrition_ttm_pct	%	Voluntary / LTM attrition, IT services; basis noted where different
it_rev_usd_mn	US$ mn	Wipro IT Services revenue; peers company revenue
cc_qoq_pct	%	Sequential revenue growth in constant currency (Wipro)
source_id	text	One or more ids from Source_Log (S01–S120)
source_type	text	Company datasheet / fact sheet / release, filing, transcript, broker research, news
note	text	Restatements, one-offs, definition changes, conflicts
qoq_margin_bps (workbook)	bps	Change from the previous consecutive quarter
qoq_util_pts (workbook)	pts	Change in utilisation from the previous consecutive quarter
wage_hike_qtr (workbook)	0/1	1 in the quarter a salary hike first hits the P&L (FY21–FY27)
usd_inr_qavg (workbook)	₹ per US$	Quarterly average of monthly Federal Reserve G.5 rates
one_off_flag (workbook)	0/1	1 where the note records a one-off, restructuring or similar charge